

Associations and trends between county-level social vulnerability and drug overdose deaths in the United States

Mark Bounthavong,^{1,2} Ariel Dong,¹ Aryana Sepassi,¹ Jamie Kneebusch¹

1. University of California, San Diego Skaggs School of Pharmacy and Pharmaceutical Sciences San Diego, CA; 2. VA Health Economics Resource Center

BACKGROUND

To investigate the association of county-level Social Vulnerability Index (SVI) changes across time on drug overdose death rates in the United States (US) from 2014 to 2020.

OBJECTIVE

To investigate the association of county-level Social Vulnerability Index (SVI) changes across time on drug overdose death rates in the United States (US) from 2014 to 2020.

METHODS

We conducted a serial cross-sectional study to evaluate the association between county-level social vulnerability with drug overdose deaths in the US from 2014 to 2020.

Our primary aim was to evaluate the associations between SVI and its four themes (Socioeconomic Status, Household Characteristics, Racial & Ethnic Minority Status, and Housing Type & Transportation) with drug overdose deaths by year (2014, 2016, 2018, and 2020).

Our secondary aim evaluated the interaction between SVI and its four themes with time on drug overdose deaths.

DISCLOSURES

No potential conflicts

MAIN FINDINGS

Drug overdose death rates increased with higher SVI and its thematic quintiles across all years studied.

In the state-adjusted model, thematic elements of SVI had similar patterns with the overall SVI in 2014, but Racial & Ethnic Minority Status and Housing Type & Transportation were also significantly associated with drug overdose death rates in 2016, 2018, and 2020.

Increasing SVI from 0 to 1 was associated with significant increases in drug overdose deaths per 100,000 persons across all years studied.

IMPLICATIONS

These increases in drug overdose death rates among vulnerable communities raise concerns about the impact of disparities and inequities on physical infrastructure and social determinants of health.

Policymakers will also need to focus their attention on other themes of SVI such as housing, which has been become a critical issue in the US.

Communities where housing instability and homelessness are critical issues will need to develop and implement strategies that are tailored to their needs.

CONCLUSIONS

Overall, counties with high levels of social vulnerability had higher rates of drug overdose deaths between 2014 and 2020.

A concerted effort between public health and local community stakeholders is needed to develop effective tailored strategies.

CONTACT

email: mbounthavong@health.ucsd.edu



Scan QR Code for copy of poster and Abstract

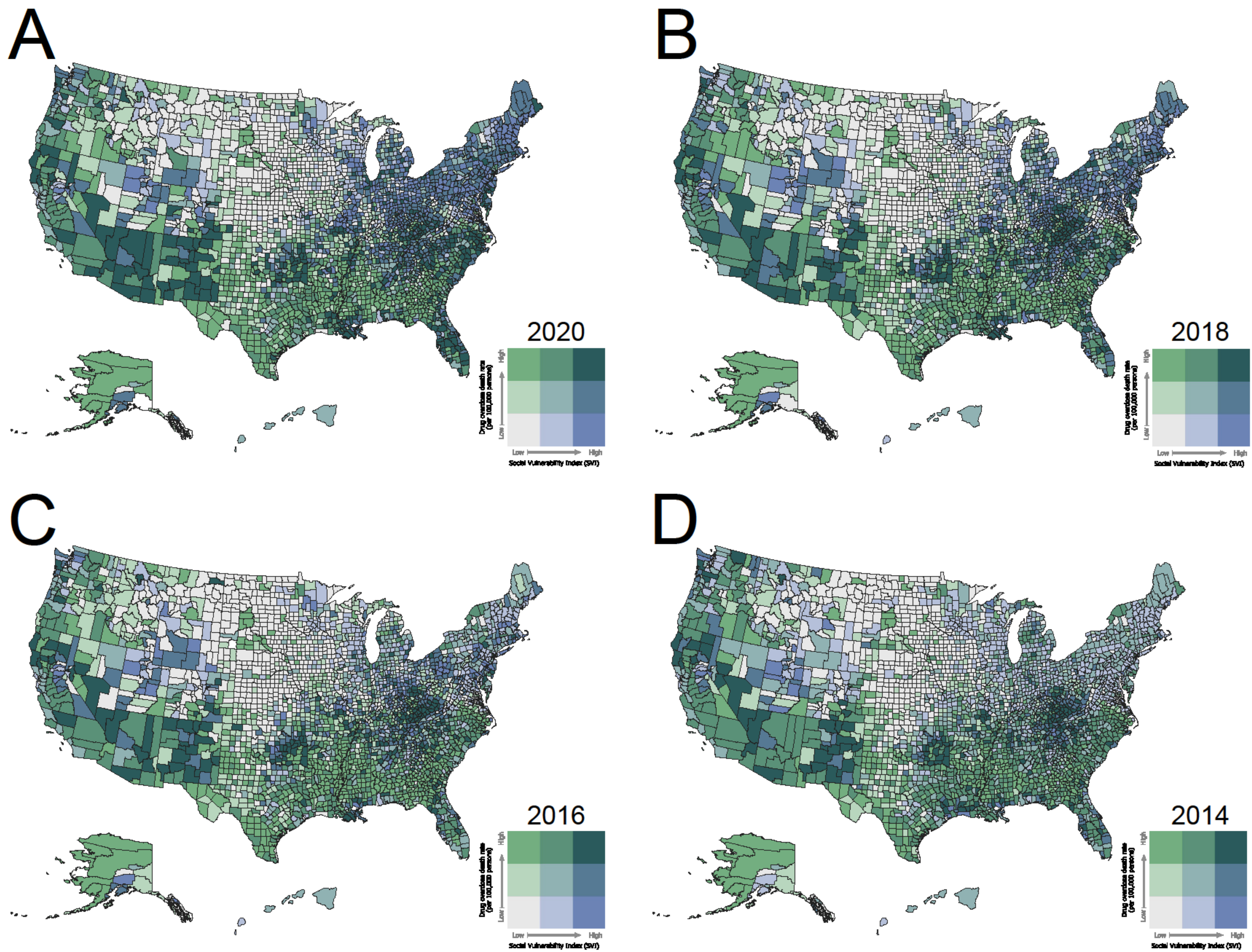


Figure: Bivariate choropleth maps of overlapping county-level social vulnerability index and drug overdose death rate.